



Pushpesh Gokul Pant

Roll No.:244156006

M.Tech

Robotics and Artificial Intelligence

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (Robotics & AI)	Indian Institute of Technology, Guwahati	8.16 (current)	2024-2026
B.Tech. (CSE)	Shri Mata Vaishno Devi University, Katra	7.47/10	2020-2024
Senior Secondary	Baluni Public School (CBSE), Dehradun	88.8%	2019
High School	The Asian Academy Sen. Sec. School (CBSE)	10/10	2017

EXPERIENCE

- Higher Education Department, Government of J&K** *Feb. 2023 – Mar. 2024*
Intern Programmer *Katra, India*
 - Built and optimized deep learning and ensemble models for medical image analysis, improving disease classification accuracy by ~12%.
 - Reviewed state-of-the-art research and implemented novel algorithms, reducing pre-processing time by 20%.
 - Documented methodologies and results, ensuring reproducibility and efficient team collaboration.
 - Tech:** Python (NumPy, Pandas, Matplotlib, Open-cv, Tensorflow, W&B, Pytorch.)

PROJECTS

- Energy Optimization of Underwater Gliders (Master Thesis Project)** *June. 2025 – Present*
Designed a multi-objective optimization framework for the Petrel-II glider, integrating dynamic modeling, environmental factors, and energy analysis.
 - Developed full kinematic-dynamic model including buoyancy, hydrodynamics, and ocean currents.
 - Extended Energy Consumption Model to cover propulsion, control, sensing, and communication loads.
 - Applied NSGA-II with surrogate modeling to optimize range and trajectory accuracy.
 - Simulations showed improved energy efficiency under variable ocean conditions.
 - Tech:** MATLAB, Python (NumPy, SciPy, Matplotlib), Optimization (NSGA-II, RBF Surrogates)
- Flare Removal** *Mar. 2025*
Re-implemented a deep learning model for lens flare removal using semi-synthetic flare-corrupted image pairs.
 - Improved generalization across real-world scenarios, achieving ~3 dB PSNR gain over baseline methods.
 - Developed custom loss functions and optimized training pipeline in PyTorch.
 - Deployed model as an API using FastAPI with Docker, and built an interactive demo with Gradio.
 - Tech:** PyTorch, scikit-learn, FastAPI, Docker, Gradio, matplotlib
- Multimodal RAG from PDFs** *Aug. 2025*
Built an end-to-end RAG system with LangChain to query PDFs containing both text and images.
 - Implemented multi-modal ingestion (OCR+image captioning) and hybrid retrieval (wtext & image embeddings).
 - Designed RAG pipeline with re-ranking and context packing, achieving grounded answers with inline citations.
 - Deployed FastAPI service with Docker, Redis caching, and observability (LangSmith) for reduced latency.
 - Tech:** Python, LangChain, FAISS, CLIP, BLIP-2, OCR, FastAPI, Docker, Redis

KEY COURSES TAKEN

- Data Structure and Algorithms, Computer Network and Security, Database Management systems, Machine Learning, Artificial Intelligence, Digital Image Processing, Operating System, Nature Inspired Algorithms

TECHNICAL SKILLS

- Programming:** Python, C++, Bash, JavaScript, SQL
- Tools & OS:** Git, Jupyter Notebook, Docker, Linux, Windows
- Libraries/Frameworks:** Pytorch, Tensorflow, FastAPI, Langgraph, Scikit-Learn, OpenCV, Numpy, Pandas
- Certifications:** AWS Cloud Foundation

POSITION OF RESPONSIBILITY

- Captain & Vice Captain(previously)**, Athletics Contingent *Nov 2024 - Present*
led the team at the **Inter IIT Sports Meet**, organized fitness camps, workshops and managed students and resources.

EXTRA-CURRICULAR

- Half Marathons (21 km)** Improved my performance from **1h 50m** at Katra (2022) to **1h 38m** at Guwahati (2025)